

```
!pip install roboflow
```

```
from roboflow import Roboflow  
rf = Roboflow(api_key="2c5zFB8IXVxMwnjF2aC8")
```

```
project = rf.workspace("machine-learning-hbxls").project("animal-  
prediction-noth1")  
dataset = project.version(1).download("folder")
```

```
Requirement already satisfied: roboflow in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (1.1.66)  
Requirement already satisfied: certifi in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (2025.4.26)  
Requirement already satisfied: idna==3.7 in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (3.7)  
Requirement already satisfied: cycler in c:\users\mubasheer\anaconda3\  
envs\moiz\lib\site-packages (from roboflow) (0.12.1)  
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\  
mubasheer\anaconda3\envs\moiz\lib\site-packages (from roboflow)  
(1.4.8)  
Requirement already satisfied: matplotlib in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (3.10.3)  
Requirement already satisfied: numpy>=1.18.5 in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (2.1.3)  
Requirement already satisfied: opencv-python-headless==4.10.0.84 in  
c:\users\mubasheer\anaconda3\envs\moiz\lib\site-packages (from  
roboflow) (4.10.0.84)  
Requirement already satisfied: Pillow>=7.1.2 in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (11.2.1)  
Requirement already satisfied: pillow-heif>=0.18.0 in c:\users\  
mubasheer\anaconda3\envs\moiz\lib\site-packages (from roboflow)  
(0.22.0)  
Requirement already satisfied: python-dateutil in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (2.9.0.post0)  
Requirement already satisfied: python-dotenv in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (1.1.1)  
Requirement already satisfied: requests in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (2.32.3)  
Requirement already satisfied: six in c:\users\mubasheer\anaconda3\  
envs\moiz\lib\site-packages (from roboflow) (1.17.0)  
Requirement already satisfied: urllib3>=1.26.6 in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (2.3.0)  
Requirement already satisfied: tqdm>=4.41.0 in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (4.67.1)  
Requirement already satisfied: PyYAML>=5.3.1 in c:\users\mubasheer\  
anaconda3\envs\moiz\lib\site-packages (from roboflow) (6.0.2)  
Requirement already satisfied: requests-toolbelt in c:\users\  
mubasheer\anaconda3\envs\moiz\lib\site-packages (from roboflow)  
(1.0.0)  
Requirement already satisfied: filetype in c:\users\mubasheer\
```

```
anaconda3\envs\moiz\lib\site-packages (from roboflow) (1.2.0)
Requirement already satisfied: colorama in c:\users\mubasheer\
anaconda3\envs\moiz\lib\site-packages (from tqdm>=4.41.0->roboflow)
(0.4.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\mubasheer\
anaconda3\envs\moiz\lib\site-packages (from matplotlib->roboflow)
(1.3.2)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\
mubasheer\anaconda3\envs\moiz\lib\site-packages (from matplotlib-
>roboflow) (4.58.4)
Requirement already satisfied: packaging>=20.0 in c:\users\mubasheer\
anaconda3\envs\moiz\lib\site-packages (from matplotlib->roboflow)
(24.2)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\mubasheer\
anaconda3\envs\moiz\lib\site-packages (from matplotlib->roboflow)
(3.2.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\
mubasheer\anaconda3\envs\moiz\lib\site-packages (from requests-
>roboflow) (3.3.2)
loading Roboflow workspace...
loading Roboflow project...
```

```
Downloading Dataset Version Zip in Animal-Prediction-1 to folder::
100%|██████████| 1105/1105 [00:01<00:00, 597.17it/s]
```

```
Extracting Dataset Version Zip to Animal-Prediction-1 in folder::
100%|██████████| 272/272 [00:00<00:00, 623.88it/s]
```

```
import tensorflow as tf
```

```
IMG_SIZE = 128
BATCH_SIZE = 32
```

```
train_ds = tf.keras.preprocessing.image_dataset_from_directory(
    "Animal-Prediction-1/train",
    image_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE
)
```

```
val_ds = tf.keras.preprocessing.image_dataset_from_directory(
    "Animal-Prediction-1/valid",
    image_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE
)
```

```
class_names = train_ds.class_names
print("Class names:", class_names)
```

Found 183 files belonging to 2 classes.
Found 26 files belonging to 2 classes.
Class names: ['Panda', 'Tiger']

```
model = tf.keras.Sequential([
    tf.keras.layers.Rescaling(1./255, input_shape=(IMG_SIZE, IMG_SIZE,
3)),
    tf.keras.layers.Conv2D(32, 3, activation='relu'),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Conv2D(64, 3, activation='relu'),
    tf.keras.layers.MaxPooling2D(),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(len(class_names), activation='softmax')
])
```

```
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

```
model.fit(train_ds, validation_data=val_ds, epochs=10)
```

Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

Epoch 1/10

6/6 ————— 6s 378ms/step - accuracy: 0.5805 - loss: 1.3862 - val_accuracy: 0.8077 - val_loss: 0.5313

Epoch 2/10

6/6 ————— 2s 276ms/step - accuracy: 0.7146 - loss: 0.6053 - val_accuracy: 0.5769 - val_loss: 0.8460

Epoch 3/10

6/6 ————— 2s 306ms/step - accuracy: 0.8042 - loss: 0.4378 - val_accuracy: 0.8077 - val_loss: 0.3922

Epoch 4/10

6/6 ————— 2s 292ms/step - accuracy: 0.8778 - loss: 0.2810 - val_accuracy: 0.8462 - val_loss: 0.3804

Epoch 5/10

6/6 ————— 2s 286ms/step - accuracy: 0.9766 - loss: 0.1598 - val_accuracy: 0.7692 - val_loss: 0.5465

Epoch 6/10

6/6 ————— 2s 283ms/step - accuracy: 0.9765 - loss: 0.1210 - val_accuracy: 0.9231 - val_loss: 0.2549

Epoch 7/10

6/6 ————— 2s 271ms/step - accuracy: 0.9843 - loss: 0.0486 - val_accuracy: 0.9231 - val_loss: 0.4004

Epoch 8/10

6/6 ————— 2s 269ms/step - accuracy: 0.9975 - loss: 0.0295 - val_accuracy: 0.9231 - val_loss: 0.2986

```
Epoch 9/10
6/6 _____ 2s 274ms/step - accuracy: 0.9975 - loss:
0.0297 - val_accuracy: 0.8846 - val_loss: 0.3710
Epoch 10/10
6/6 _____ 2s 273ms/step - accuracy: 0.9975 - loss:
0.0190 - val_accuracy: 0.9615 - val_loss: 0.2674

<keras.src.callbacks.history.History at 0x13f7fe3bc40>

model.save("my_animal_model.keras")

model = tf.keras.models.load_model("my_animal_model.keras")

Skipping variable loading for optimizer 'rmsprop', because it has 10
variables whereas the saved optimizer has 18 variables.
```

```

import tensorflow as tf
#saved model
model = tf.keras.models.load_model("my_animal_model.keras",
compile=False)

import os
from tensorflow.keras.preprocessing import image
import numpy as np
import matplotlib.pyplot as plt

img_path = "D:\moiz\dataset\Panda\Panda_1_3.jpg"

class_names = os.listdir("Animal-Prediction-1/train")
class_names.sort()

# Load image and preprocess
img = image.load_img(img_path, target_size=(128, 128))
img_array = image.img_to_array(img) / 255.0
img_array = np.expand_dims(img_array, axis=0)

# Predict
prediction = model.predict(img_array)
predicted_index = np.argmax(prediction[0])
confidence = prediction[0][predicted_index]

1/1 ————— 0s 428ms/step

# Show result
plt.imshow(img)
plt.title(f"Predicted: {class_names[predicted_index]}
({confidence*100:.2f}%)")
plt.axis('off')
plt.show()

```

Predicted: Panda (99.63%)



```
img_path = "D:\moiz\dataset\Tiger\Tiger_1.jpg"

class_names = os.listdir("Animal-Prediction-1/train")
class_names.sort()

# Load image and preprocess
img = image.load_img(img_path, target_size=(128, 128))
img_array = image.img_to_array(img) / 255.0
img_array = np.expand_dims(img_array, axis=0)

# Predict
prediction = model.predict(img_array)
predicted_index = np.argmax(prediction[0])
confidence = prediction[0][predicted_index]

# Show result
plt.imshow(img)
plt.title(f"Predicted: {class_names[predicted_index]} ({confidence*100:.2f}%)")
plt.axis('off')
plt.show()

1/1 ————— 0s 142ms/step
```

Predicted: Panda (99.59%)



```
img_path = "D:\moiz\dataset\Tiger\Tiger_6_2.jpg"

class_names = os.listdir("Animal-Prediction-1/train")
class_names.sort()

# Load image and preprocess
img = image.load_img(img_path, target_size=(128, 128))
img_array = image.img_to_array(img) / 255.0
img_array = np.expand_dims(img_array, axis=0)

# Predict
prediction = model.predict(img_array)
predicted_index = np.argmax(prediction[0])
confidence = prediction[0][predicted_index]

# Show result
plt.imshow(img)
plt.title(f"Predicted: {class_names[predicted_index]} ({confidence*100:.2f}%)")
plt.axis('off')
plt.show()

1/1 ————— 0s 104ms/step
```


Predicted: Panda (99.60%)



```
img_path = "D:\moiz\dataset\Panda\Panda_2_3.jpg"

class_names = os.listdir("Animal-Prediction-1/train")
class_names.sort()

# Load image and preprocess
img = image.load_img(img_path, target_size=(128, 128))
img_array = image.img_to_array(img) / 255.0
img_array = np.expand_dims(img_array, axis=0)

# Predict
prediction = model.predict(img_array)
predicted_index = np.argmax(prediction[0])
confidence = prediction[0][predicted_index]

# Show result
plt.imshow(img)
plt.title(f"Predicted: {class_names[predicted_index]} ({confidence*100:.2f}%)")
plt.axis('off')
plt.show()

1/1 ————— 0s 120ms/step
```


Predicted: Panda (99.60%)

